Brendan J. Clark, Ph.D.

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Curriculum Vitae · November 2026

EDUCATION

Jan. 2025 **Ph.D., Atmospheric Science** · Rutgers University, New Brunswick
Advisor: Prof. Alan Robock
Dissertation: *Stratospheric Aerosol Climate Intervention Implications for Global Agriculture*

Nov. 2022 **M.S., Atmospheric Science** · Rutgers University, New Brunswick
Advisor: Prof. Alan Robock
Thesis: *Optimal climate intervention scenarios for crop production vary by nation*

May 2020 **B.S., Environmental Science** · University of Massachusetts, Amherst
Certificate in GIS; minor in Natural Resource Conservation

ACADEMIC EXPERIENCE

2025–present **Research Associate** · Cornell University, Ithaca, NY
Integrating eddy covariance, aerosol observations, radiative transfer modelling, and satellite imagery to quantify radiative effects of wildfire smoke on crop and forest growth. Collaborated with CarbonPlan to build a Python pipeline for statistical downscaling of climate simulations and public data access via AWS. Developed machine learning models to assess climate change impacts on forest carbon and benchmarked against Earth system model simulations.

2020–2025 **Research Assistant** · Rutgers University, New Brunswick, NJ
Developed CESM model code and ran simulations on the Derecho supercomputer using FORTRAN, shell scripting, and Slurm. Downscaled and bias-corrected climate model output and ran global crop model simulations to assess climate change and geoengineering impacts on crop production and nutritional quality.

2019–2020 **Research Intern** · University of Massachusetts, Amherst, MA
Analyzed trace gas emissions from shale samples using gas chromatography to investigate secondary methane and nitrous oxide emissions associated with hydraulic fracturing.

PUBLICATIONS 11 total (6 published)

11. **B. Clark**, B. Kravitz, D. Visoni, M. Lerdau. Analogs for stratospheric aerosol injection suggest modest, positive radiative effects on plants. *(in preparation)*.
10. **B. Clark**, C. M. Zarakas, O. S. Chegwidden, T. S. Egbebiyi, A. Banihirwe, N. R. Hagen, W. R. Lee, M. Henry, H. Sipra, R. Geiger, and D. Visioni. Global downscaled climate projections of stratospheric aerosol injection at 0.25° daily resolution. *(in review)*.
9. C. Wang, D. Visioni, **B. Clark**, Z. M. Labe, B. Jong, W. Lee, H. Hirasawa. Inter-model differences in monsoon response under SRM are greater than multi-scenario SRM differences. *(in review)*.
8. **B. Clark**, D. Visioni. Canopy structure amplifies the role of radiation in vegetation responses of stratospheric aerosol injection. *(in review)*.
7. N. Grant, A. Robock, L. Xia, J. Kiniry, and **B. Clark**. Can solar climate intervention save coffee and chocolate from climate change? *(in review)*.
6. **B. Clark**, D. Visioni, S. Kothari, M. Lerdau. Land models likely underestimate the impact of future atmospheric dryness on European tree growth. *Geophysical Research Letters* 53, e2026GL122759 (2026). 10.1029/2026GL122759

5. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, G. Hoogenboom, J. Jaegermeyr. Maize yield changes under sulfate aerosol climate intervention using three global gridded crop models. *Earth's Future* 13, e2024EF005269 (2025). 10.1029/2024EF005269
4. N. Grant, A. Robock, L. Xia, J. Singh, **B. Clark**. Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future* 13, e2024EF005262 (2025). 10.1029/2024EF005262
3. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, J. Jaegermeyr. Stratospheric aerosol climate intervention could reduce crop nutritional value. *Environmental Research Letters* 20, 114083 (2025). 10.1088/1748-9326/ae1151
2. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. Optimal climate intervention scenarios for crop production vary by nation. *Nature Food* 4, 902–911 (2023). 10.1038/s43016-023-00853-3
1. F. Bowlick, **B. Clark**, et al. Understanding geocomputation education: A survey and syllabi informed review. *Research in Geographic Education* 23, 20–51 (2022).

FUNDING AND AWARDS

2026	**Gordon Research Conference Travel Support Award** ($2,500)
2026	**CESM Workshop Travel Award** ($3,000)
2025	**Climate Intervention Network Travel Award** ($1,000)
2025	**NSF NCAR Exploratory Allocation Award** (500,000 CPU hours)
2025	**GeoMIP Annual Meeting Travel Funding Award** ($3,500)
2024	**Rutgers Atmospheric Science Student Travel Support Award** ($1,500)
2023	**Rutgers Atmospheric Science Student Travel Support Award** ($1,200)
2022	**Rutgers Climate Institute Student Travel Support Award** ($1,000)
2022	**NSF NCAR University Small Allocation Award** (200,000 CPU hours)
2019	**Governor's Award in Recognition of Environmental Stewardship**
2016	**John and Abigail Adams Scholarship Award** ($7,000)
2016	**Stanley Z. Koplik Certificate of Mastery with Distinction Award**

TEACHING EXPERIENCE

Spring 2026	**EAS 5555: Theory and Practice of Earth System Modeling** · Cornell University Guest lecture
Spring 2026	**01:090:297: Climate Intervention: Where Science Meets Ethics** · Rutgers University Guest lecture
Fall 2024	**16:107:544: Modeling Climate Change** · Rutgers University Guest lecture
Fall 2019	**NRC 585: Introduction to GIS** · University of Massachusetts, Amherst Teaching Assistant
2018–2019	**Algonquin Regional High School** Substitute teacher; occasional scientific lectures to students

CONFERENCE PRESENTATIONS first author only; † denotes invited

July 2026	**B. Clark**, D. Visioni, S. Kothari, M. Lerdau. "Land models likely underestimate the effect of atmospheric dryness on tree growth." Talk, Ecological Society of America, Salt Lake City, UT.
June 2026	**B. Clark**. "Land models likely underestimate benefits to natural vegetation under stratospheric aerosol injection." Talk, Gordon Research Conference, Newry, ME. †

June 2026	**B. Clark**, D. Visioni, S. Kothari, M. Lerdau. "Land models likely underestimate the effect of atmospheric dryness on tree growth." Talk, CESM Working Group Meeting, Boulder, CO.
March 2026	**B. Clark**, D. Visioni, S. Kothari, M. Lerdau. "Land models likely underestimate the effect of atmospheric dryness on European tree growth." Talk, Coupled Model Intercomparison Project Community Workshop, Kyoto, Japan.
March 2026	**B. Clark**, D. Visioni. "Altered sunlight under stratospheric aerosol injection would have heterogeneous impacts on vegetation." Poster, GeoMIP Annual Meeting, Tokyo, Japan.
Dec. 2025	**B. Clark**, D. Visioni, S. Kothari, M. Lerdau. "Land models likely underestimate the effect of atmospheric dryness on European tree growth." Talk, American Geophysical Union Meeting, New Orleans, LA.
Dec. 2025	**B. Clark**, D. Visioni, S. Kothari, M. Lerdau. "Using dendrometer data to inform projections of carbon storage under solar radiation modification." Poster, American Geophysical Union Meeting, New Orleans, LA.
May 2025	**B. Clark**. "Statistically downscaled and bias-adjusted ARISE-SAI and SSP2-4.5 scenarios." Poster, DEGREES Global Forum, Cape Town, South Africa.
Dec. 2024	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Jaegermeyr, J. Guarin. "Stratospheric aerosol climate intervention could reduce the nutritional value of maize and rice." Talk, American Geophysical Union Meeting, Washington, D.C.
July 2024	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Jaegermeyr, J. Guarin. "Sulfate aerosol climate intervention impacts on maize yield and protein in three global gridded crop models." Poster, GeoMIP Annual Meeting, Cornell University.
Feb. 2024	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Jaegermeyr, J. Guarin. "Stratospheric aerosol climate intervention could negatively impact crop nutritional quality." Talk, Gordon Research Seminar, Newry, ME.
Feb. 2024	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Jaegermeyr, J. Guarin. "Stratospheric aerosol climate intervention impacts on crop protein content." Poster, Gordon Research Conference, Newry, ME.
Dec. 2023	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Jaegermeyr, J. Guarin. "Stratospheric aerosol climate intervention could negatively impact crop nutritional quality." Talk, American Geophysical Union Meeting, San Francisco, CA.
July 2023	**B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. "The optimal climate intervention scenario for crop production varies by nation." Talk, Exeter, UK. †
July 2023	**B. Clark**, A. Robock, L. Xia, S. Rabin, J. Singh, N. Grant. "Rutgers Impact Studies of Climate Intervention (RISCI) laboratory group overview." Poster, GeoMIP Annual Meeting, Exeter, UK.
June 2023	**B. Clark**, A. Robock, L. Xia. "A proposal for a multi-crop model assessment of stratospheric aerosol climate intervention." Talk, AgMIP-GGCMI Meeting, Columbia University.
Dec. 2022	**B. Clark**, A. Robock, L. Xia. "Can crop production be used as a metric to design climate intervention?" Poster, American Geophysical Union Meeting, Chicago, IL.
June 2022	**B. Clark**, A. Robock, L. Xia. "Impacts on crop production from stratospheric aerosol climate intervention: a multi-scenario overview." Poster, Gordon Research Conference, Newry, ME.
May 2022	**B. Clark**, A. Robock, L. Xia. "Crop impacts from stratospheric aerosol injection: a multi-scenario overview." Talk, University of Potsdam, Germany. †

Feb. 2022	**B. Clark**, A. Robock, L. Xia. "Discrepancies between fully coupled and offline CLM5 crop simulations." Talk, NCAR Land Modeling Working Group Meeting, Boulder, CO.
Dec. 2021	**B. Clark**, A. Robock, L. Xia. "Impacts on crop production from stratospheric aerosol injection." Poster, American Geophysical Union Meeting, New Orleans, LA.
Oct. 2021	**B. Clark**, A. Robock, L. Xia. "Depicting information and remembering your audience: impacts on crop production from stratospheric aerosol climate intervention." Talk, Climate Engineering in Context Conference, University of Potsdam, Germany.

TECHNICAL SKILLS

Programming	Python, R, MATLAB, Fortran, JavaScript, HTML, bash shell scripting
Python Libraries	Data: xarray, numpy, pandas, scipy, dask Visualization: matplotlib, seaborn, plotly Machine learning / statistics: scikit-learn, statsmodels
Computing	High-performance computing (NCAR Casper/Derecho, university clusters); cloud computing (AWS); parallelization with Python multiprocessing and Slurm scheduling
Tools	Git and GitHub; Jupyter notebooks; Google Colab

SERVICE

Journal reviewer	Geophysical Research Letters; Environmental Research Letters; Earth's Future; Atmospheric Chemistry and Physics; Journal of Geophysical Research Atmospheres; npj Climate and Atmospheric Science; Environmental Research Climate; PLOS Climate
Grant Reviewing	The Degrees Initiative — grant proposal reviewer Reflective — grant proposal reviewer
Organizing	AGU 2026 Session Convener, "Advances in Climate Engineering Science"
Outreach	Physicists Coalition for Nuclear Threat Reduction Science Policy and Advocacy at Rutgers

PROFESSIONAL AFFILIATIONS

American Geophysical Union · Ecological Society of America · Agricultural Modeling Intercomparison Project · Global Gridded Crop Modeling Intercomparison Project · Geoengineering Modeling Intercomparison Project · Climate Intervention Biology Working Group · Inter-Sectoral Impact Model Intercomparison Project

SELECTED MEDIA COVERAGE

"What happens to our food supply when we dim the sun?" *Climate Chat Podcast*
"A blind spot in climate models: Slower tree growth in a warming world" *The Bulletin of Atomic Scientists*
"Getting warmer: Slower forest growth means less carbon storage" *Cornell Chronicle*
"Blocking the sun may fight climate change but kill our crops" *The Daily Beast*
"Climate intervention technologies may create winners and losers in world food supply" *Rutgers Today*